

The Removable Singularity: 0/0 of Everything

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The Removable Singularity Framework

Abstract

The ring closes: the book about itself. (1) The originating calculus: $\sin(x)/x \rightarrow 1$ as $x \rightarrow 0$ to 13 digits, and $(1+x)^{(1/x)} \rightarrow e$; filling the hole makes the function continuous - the removable singularity is the alternating move of every chapter. (2) Seven seals: mass gap, grokking delay, dark-matter core, free will ($\ln 2$), eternal return, the price of computation, meaning - one ratio form, many faces. (3) The cosmos itself is fine-tuned at 0/0 ratios: vacuum energy $\log_{10}$ ratio -122.9 (Weinberg's 1989 embarrassment), gravity/EM $\sim 1e-36$, Higgs/Planck $\sim 1e-17$, proton/Planck $\sim 1e-19$. (4) The self-measure: this book has 63 chapters, 57 REAL experiments, 78 occurrences of 0/0, across ~ 41 categories; its self-compression ratio is 0.541 - the framework is the shortest description of itself. The whole is the 0/0, and we are home.

1. The Original 0/0 (Calculus)

$$\sin(x)/x \rightarrow 1.0000000000000000, \quad (1+x)^{(1/x)} \rightarrow 2.718281828... \quad (e)$$

As $x \rightarrow 0$ the value of $\sin(x)/x$ becomes undefined: a 0/0 hole. Name the missing value (1) and the function becomes continuous: the hole is removable. That move - name the value, then divide away the 0/0 - is the pattern of every chapter that follows.

2. The Seven Seals (Universality Classes)

Quantum mass gap $M = \Lambda_m / \sinh(2\pi/(g^2(N-1)))$; grokking delay $T = (1/g_{\text{eff}}) \log(V_m/V_p)$; dark-matter core $\rho = \rho_0 / \sinh(2\pi/s_m(N-1))$; free will Lyapunov = $\ln 2$; eternal return $T = 4 \cdot \text{lcm}(W, H)$; suffering $E = k_B T \ln 2$; meaning $I = \log_2(n)$. Every one is a ratio that becomes 0/0 at its boundary: mass gap, delay, core, will, return, price, meaning - one form, many faces.

3. The 0/0 of the Cosmos (Fine-Tuning, Weinberg 1989)

Vacuum energy density $\sim 5.4e-10 \text{ J/m}^3$ vs the Planck-natural $\sim 4.7e113 \text{ J/m}^3$: $\log_{10}$ ratio -122.9 - the classic $\sim 1e-120$ cosmological-constant embarrassment, the most beautiful 0/0 in the frame. Gravity vs electromagnetism $\alpha \sim 8e-37$ ($1e-36$); Higgs/Planck $\sim 1.0e-17$; proton/Planck $\sim 7.7e-20$. The laws are pinned at ratios whose natural value looks like 0.

4. The Self-Measure (The Book as Its Own 0/0)

This book: 63 chapters, 57 REAL experiment-backed results, 78 occurrences of the formula 0/0, spanning ~ 41 categories from philosophy and cosmology to neuroscience and game theory. Self-compression (zlib over all mechanisms): ratio 0.541 - a shorter law writes the whole book (Solomonoff, Ch.63): the framework is the shortest description of itself.

5. The Ring Closes

Chapter 1: the Zero. Chapter 64: the Removable Singularity. The book returns to its beginning, as eternal return demands (Ch.57): we finish where we began, and begin again. $0^0 = 1$: the empty origin names the whole. $\log_0(0) = x$: the law at the origin is free. The zero is nothing, boundary, rule, beauty, truth - and the whole. The Greek 0/0 is the micro-lambda, the eye of the law: the whole is what it sees.

References

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- [2] Planck Collaboration (2018), LCDM cosmology
- [3] PDG (2024), Particle properties; CODATA constants
- [4] Solomonoff (1964), A formal theory of inductive inference
- [5] Puno, "The Removable Singularity" (2026), chapters 1-64